Endogeneity, Instrumental Variables

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Example of Endogeneity

Consider a simple model:

$$\text{Income} = \alpha + \beta \cdot \text{Education} + \epsilon$$

What does this model imply?

- Education (X) is expected to positively affect Income (Y).
- The error term (ϵ) captures other factors affecting Income, like ability, motivation, or family background.
- These factors (ϵ) might also influence Education (X).

Problem:

- If ability, which affects both Income and Education, is omitted from the model, then Education (X) and the error term (ϵ) are correlated.
- This correlation between X and ϵ leads to endogeneity, resulting in biased estimates of β.

Endogeneity

$$Y = \alpha + \beta X + \epsilon$$

1. X causes Y
2. ϵ causes Y
3. ϵ does not cause X
4. Y does not cause X

If X is correlated with ϵ, X is said to be an endogenous explanatory variable.

Common Types of Endogeneity

- Omitted variables
  - $Y = \alpha + \beta X + \gamma Z + u = \alpha + \beta X + \epsilon$
  - variables X and Z are correlated
- Measurement error
  - we observe $X = X^* + u$
  - $Y = \alpha + \beta X^* + \nu = \alpha + \beta X - \beta u + \nu = \alpha + \beta X + \epsilon$
- Self-selection
  - Participation is not determined randomly
  - $E[\epsilon | participation]$
- Simultaneity
  - e.g. demand and supply

Example: Simultaneity in the Apartment Market

Scenario:

- Consider the rental apartment market in a city.
- Rent price (P) and quantity of apartments rented (Q) are determined by both demand and supply.

Demand Side:

- The demand for apartments (Q_d) decreases as rent prices increase.
- Example Demand Equation:

$$Q_d = \alpha + \beta P + \epsilon$$

- ϵ represents factors like population growth, income growth, or a new company moving to the city.

Supply Side:

- The supply of apartments (Q_s) increases as rent prices increase.
- Example Supply Equation:

$$Q_s = \gamma + \delta P + \nu$$

- ν include, e.g., construction costs or regulatory constraints.

Simultaneity in the Apartment Market

Market Equilibrium:

- At equilibrium, the quantity of apartments demanded equals the quantity supplied:

$$Q = Q_d = Q_s$$

- Both Q and P are determined together at equilibrium.

Demand Shock Example:

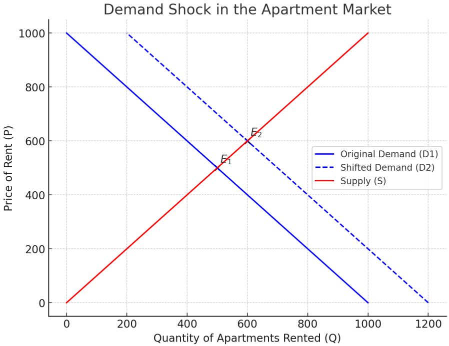
- A large company moves its headquarters to the city (ϵ increases).
- This increases demand, shifting the demand curve.
- As a result, both the quantity of apartments rented (Q) and the price of rent (P) increase.

Key Point:

- The shift in demand due to ϵ causes a correlation between P and ϵ, leading to simultaneity.
- This correlation can bias the estimation of demand and supply functions.

Demand Shock in the Apartment Market

- A large company moves to the city, increasing demand for apartments.
- The demand curve shifts right, raising both rent (P_2) and quantity rented (Q_2).
- New equilibrium (E_2) shows simultaneous changes in P and Q.



P and ϵ are correlated, leading to simultaneity and potential bias.

Instrumental Variables

$$Y = \alpha + \beta X + \epsilon, \text{ where } Cov(X, \epsilon) \neq 0 \tag{1}$$

Suppose that we have an observable variable z that satisfies these two assumptions:

1. z is uncorrelated with ϵ: $Cov(z, \epsilon) = 0$
2. z is correlated with X: $Cov(z, X) \neq 0$

Then we call z an instrumental variable for X

- (1) is called the structural equation
- an endogenous variable (Y) in terms of another endogenous variable (X)

Intuition of Instrumental Variables

$$
\begin{tikzcd}
 & X \arrow[dl, dash] \arrow[dr, dash] & \\
X_B & & X_G
\end{tikzcd}
$$

$$\text{Cov}(X_B, \epsilon) \neq 0 \qquad \text{Cov}(X_G, \epsilon) = 0$$

$$Y = \beta_0 + \beta_1 X_G + \epsilon,$$
$$\text{where } X = X_G + X_B \text{ and}$$
$$X_G = \gamma_0 + \gamma_1 z$$

- X_B **(Bad X):** The endogenous component.
- X_G **(Good X):** The exogenous component ($\text{Cov}(X_G, \epsilon) = 0$) is given by the instrument variable z.

$Cov(z, X) \neq 0$

- z is correlated with X can be tested:

$$X = X_G + X_B = \gamma_0 + \gamma_1 z + \nu \quad (2)$$

- If γ_1 is significant, then we can be fairly confident that $Cov(z, X) \neq 0$
- (2) is an example of a reduced form equation
  - write an endogenous variable in terms of exogenous variables

Example: Estimating the causal effect of skipping classes on final exam score

$$score = \beta_0 + \beta_1 skipped + u$$

- *skipped* might be correlated with other factors in u
  - more able and highly motivated students might miss fewer classes
- IV candidate: *distance* between living quarters and campus
  - Bad weather, oversleeping etc. can cause students to miss classes
    - $skipped = \gamma_0 + \gamma_1 distance + \nu$
  - Is the sign of γ_1 important?
  - Is *distance* uncorrelated with u?

Multiple regression model

$$Y = \beta_0 + \beta_1 X + \beta_2 z_1 + \cdots + \beta_k z_{k-1} + \epsilon \tag{3}$$

- X: endogenous explanatory variable
- z_k: exogenous variable not in (3), this is our instrumental variable (IV).

$$X = \pi_0 + \pi_1 z_1 + \cdots + \pi_{k-1} z_{k-1} + \pi_k z_k + v \tag{4}$$

- we require $\pi_k \neq 0$

Two stage least squares

A simple example

$$Y = \beta_0 + \beta_1 X + \beta_2 z_1 + \epsilon \qquad \text{(structural)}$$

$$X = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v \qquad \text{(reduced form)}$$

- Think of reduced form as breaking X into two parts:

  $\pi_0 + \pi_1 z_1 + \pi_2 z_2$ uncorrelated with ϵ (good X),

  v correlated with ϵ (bad X)

- (1st stage) Estimate the reduced form by OLS and obtain the fitted values (estimated good X):

$$\hat{X} = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \hat{\pi}_2 z_2$$

- (2nd stage) Estimate the structural equation using OLS and the fitted values

$$Y = \beta_0 + \beta_1 \hat{X} + \beta_2 z_1 + \epsilon$$

Intuition of 2SLS

- $\hat{X}$ is the estimate of $X_G = \pi_0 + \pi_1 z_1 + \pi_2 z_2$
- X_G is uncorrelated with ϵ
- 2SLS first "purges" X of its correlation with ϵ before doing OLS
- Recall $X = X_G + v$

$$
\begin{aligned}
Y &= \beta_0 + \beta_1 X + \beta_2 z_1 + \epsilon \\
&= \beta_0 + \beta_1 X_G + \beta_2 z_1 + \epsilon + \beta_1 v
\end{aligned}
$$

Good empirical practice

- Look at the reduced forms
- Report the OLS results for the structural model
- Report the 2SLS results

Tests

Weak Instruments Low correlation between z and X

H_0 : the IV is weak (reject)

Hausman Test of endogeneity of X

H_0 : X is exogenous (reject)

Sargan Only if you have more IVs

H_0 : all IVs are exogenous (do not reject)

Weak Instruments Test

- Our example:

$$Y = \beta_0 + \beta_1 X + \beta_2 z_1 + \epsilon \qquad \text{(structural)}$$
$$X = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v \qquad \text{(reduced form)}$$

- Tests if the slope coefficient π_2 for instrument z_2 is zero.

$$\hat{X} = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \hat{\pi}_2 z_2$$

Hausman Test Overview

- The Hausman test is used to determine whether an explanatory variable in your regression model is endogenous, meaning that it is correlated with the error term.
- This test compares the estimates from a two-stage least squares (2SLS) regression with those from an ordinary least squares (OLS) regression.

Hausman Test: IV Regression and Hypotheses

Here's how it works:
Example:

$$y = \beta_0 + \beta_1 X + \beta_2 z + \epsilon$$

Here, X is the potentially endogenous variable, and z is exogenous.

1. Run the OLS regression: Estimate your model using OLS.

2. Run the IV regression (2SLS): Estimate the same model using an instrumental variable for X, producing 2SLS estimates.

3. Formulate the null and alternative hypotheses:

- Null hypothesis (H_0): The β_1-estimates of OLS and 2SLS are the same, implying that X is exogenous.
- Alternative hypothesis (H_1): The β_1-estimates of OLS and 2SLS are different, indicating X is endogenous.

Hausman Test: Calculate the Test Statistic

4. Calculate the test statistic: The test compares the difference between the OLS and IV estimates. The test statistic is computed as:

$$H = \frac{(\hat{\beta}_{OLS} - \hat{\beta}_{IV})^2}{Var(\hat{\beta}_{OLS}) - Var(\hat{\beta}_{IV})}$$

where $\hat{\beta}_{OLS}$ and $\hat{\beta}_{IV}$ are the 2SLS and OLS β_1-estimates, respectively.

This test statistic follows a chi-square distribution with degrees of freedom equal to the number of parameters tested (usually the number of endogenous regressors).

Hausman Test: Interpret the Results

5. Interpret the results:

- If the test statistic is significantly large, you reject the null hypothesis, suggesting that X is endogenous.
- If the test statistic is not significant, you do not reject the null hypothesis, implying X is exogenous, and OLS estimates are consistent.

Sargan test for overidentifying restrictions

Consider two instrumental variables z_3 and z_4.

$$Y = \beta_0 + \beta_1 X + \beta_2 z_1 + \beta_3 z_2 + \epsilon \tag{5}$$

- We have more instruments than we need
- If all instruments are exogenous, the 2SLS residuals should be uncorrelated with the instruments

1. Estimate the structural equation by 2SLS and obtain the residuals $\hat{\epsilon}$
2. Regress $\hat{\epsilon}$ on all exogenous variables. Obtain the R-squared, R_1^2
    - This R_1^2 is hopefully low
3. Test statistic is $nR_1^2 \sim \chi_q^2$, where $q = \#IVs - \#endo$, n is the number of sample points

Instrumental Variables: Earning-schooling

- Consider the following OLS regression:

$$y = \beta_1 x + \underbrace{(\beta_2 m + e)}_{u}$$

where y = log of earnings
x = years of schooling
m = learning ability
e, u = error terms

- We want to measure how an *exogenous* change in x affects y (i.e.,β_1) :
  - there is association between x and u
  - as a consequence, there is both a direct effect via $\beta_1 x$ and an indirect effect via u affecting x which in turn effects y
  - thus, β_1 is biased

Earning-schooling, Cont'd

- Definition of an instrument:

$$\begin{tikzcd} z \arrow[r] & x \arrow[r] & y \\ & u \arrow[u] \arrow[ur] & \end{tikzcd}$$

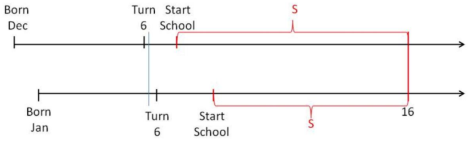
- A variable *z* is called an instrument or instrumental variable for the regressor *x* in the scalar regression model if

  (1) *z* is uncorrelated with error *u* (i.e. cov(*z*, *u*) = 0)

  (2) *z* is correlated with the regressor *x* (i.e. cov(*z*, *x*) ≠ 0)

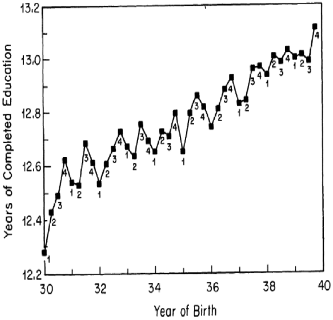
Earning-schooling, Cont'd

- Selecting the instrument requires domain knowledge:
  - In the returns to education literature Angrist and Krueger (1991) used *quarter of birth* as an instrument al variable for schooling
  - In the US, you could drop out of school once you turned 16
  - Children have different ages when they start school and thus different lengths of schooling at the time they turn 16 when they can potentially drop out

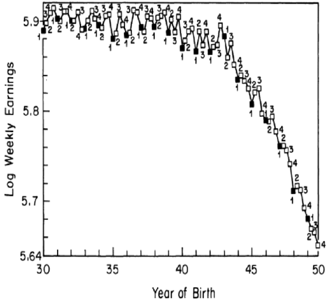


Earning-schooling, Cont'd

- First condition: cov(z,x)≠0
- Men born earlier in the year have lower schooling. This indicates that there is a first stage.



- Second condition: cov(z,u)=0
- Difference in schooling due to different quarter of birth translates into different earning with a fixed age. This indicates that there is a second stage.



Earning-schooling, Cont'd

Outcome variable	Birth cohort	Mean	Quarter-of-birth effect[a] I	II	III	F-test[b] [P-value]
Total years of education	1930–1939	12.79	−0.124 (0.017)	−0.086 (0.017)	−0.015 (0.016)	24.9 [0.0001]
	1940–1949	13.56	−0.085 (0.012)	−0.035 (0.012)	−0.017 (0.011)	18.6 [0.0001]
High school graduate	1930–1939	0.77	−0.019 (0.002)	−0.020 (0.002)	−0.004 (0.002)	46.4 [0.0001]
	1940–1949	0.86	−0.015 (0.001)	−0.012 (0.001)	−0.002 (0.001)	54.4 [0.0001]
Years of educ. for high school graduates	1930–1939	13.99	−0.004 (0.014)	0.051 (0.014)	0.012 (0.014)	5.9 [0.0006]
	1940–1949	14.28	0.005 (0.011)	0.043 (0.011)	−0.003 (0.010)	7.8 [0.0017]
College graduate	1930–1939	0.24	−0.005 (0.002)	0.003 (0.002)	0.002 (0.002)	5.0 [0.0021]
	1940–1949	0.30	−0.003 (0.002)	0.004 (0.002)	0.000 (0.002)	5.0 [0.0018]

- The table reports estimates of each quarter of birth (main) effect relative to the fourth quarter
- The F-tests indicate that after removing trends, the small within-in-year-of-birth differences in average years of education are highly statistically significant
  - F-statistics is for a test of the hypothesis that the quarter-of-birth dummies jointly have no effects
- Quarter of birth is a strong predictor of total years of education
- Reassuring quarter of birth does not affect the probability of graduating from college

Earning-schooling, Cont'd

- Estimation results with quarter of birth IV dummies:

Independent variable	(1) OLS	(2) TSLS	(3) OLS	(4) TSLS
Years of education	0.0711 (0.0003)	0.0891 (0.0161)	0.0711 (0.0003)	0.0760 (0.0290)
Race (1 = black)	—	—	—	—
SMSA (1 = center city)	—	—	—	—
Married (1 = married)	—	—	—	—
9 Year-of-birth dummies	Yes	Yes	Yes	Yes
8 Region-of-residence dummies	No	No	No	No
Age	—	—	−0.0772 (0.0621)	−0.0801 (0.0645)
Age-squared	—	—	0.0008 (0.0007)	0.0008 (0.0007)
χ^2 [dof]	—	25.4 [29]	—	23.1 [27]

- The 2SLS estimates do not equal the OLS estimates in this over-identified model, and the standard errors from OLS regression is much smaller than those from 2SLS regression